

Choice of POD Metric and Stabilization of High Modes in Case-2 LSPG: Mathematical Formulation and Numerical Results

Project_YvonMaday

May 21, 2026

1 Purpose of this note

This note presents, in a single document, the mathematical formulation of the three metric-based proposals and the numerical evidence obtained in the current test campaign. The presentation follows the same didactic style used in the previous theoretical note: first the model and reduced formulations, then the proposals, and finally the numerical interpretation.

The reference test point is

$$\boldsymbol{\mu} = (\mu_1, \mu_2) = (4.56, 0.019),$$

with reduced dimensions

$$n_{\text{tot}} = 151, \quad n = 10, \quad \bar{n} = n_{\text{tot}} - n = 141.$$

2 Context and notation

The full-order model is written in residual form as

$$\mathbf{R}(\mathbf{u}(\boldsymbol{\mu}); \boldsymbol{\mu}) = \mathbf{0}.$$

For each parameter sample, a high-fidelity solver provides states

$$\mathbf{u}_h(\boldsymbol{\mu}_i) \in \mathbb{R}^N, \quad i = 1, \dots, N_{\text{snap}},$$

from which a snapshot matrix is assembled:

$$\mathbf{S} = [\mathbf{u}_h(\boldsymbol{\mu}_1), \dots, \mathbf{u}_h(\boldsymbol{\mu}_{N_{\text{snap}}})] \in \mathbb{R}^{N \times N_{\text{snap}}}.$$

The reduced basis is split as

$$\mathbf{V}_{\text{tot}} = [\mathbf{V} \ \overline{\mathbf{V}}], \quad \mathbf{q}_N = \begin{bmatrix} \mathbf{q} \\ \overline{\mathbf{q}} \end{bmatrix},$$

with affine reconstruction

$$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \overline{\mathbf{V}}\overline{\mathbf{q}}.$$

3 Discrete residual and reduced objectives

At one implicit time step t_{k+1} , for fixed parameter $\boldsymbol{\mu}$, with previous state $\mathbf{w}_k$, the discrete residual is

$$\mathbf{r}^{k+1}(\mathbf{w}; \boldsymbol{\mu}) = \mathbf{0},$$

and the Jacobian is

$$\mathbf{J}^{k+1}(\mathbf{w}; \boldsymbol{\mu}) = \frac{\partial \mathbf{r}^{k+1}}{\partial \mathbf{w}}(\mathbf{w}; \boldsymbol{\mu}).$$

Linear reduced reference problem

The linear reduced trajectory is obtained by solving, at each step,

$$\mathbf{q}_{N,k+1}^{lin} = \arg \min_{\mathbf{q}_N \in \mathbb{R}^{n_{tot}}} \frac{1}{2} \left\| \mathbf{r}^{k+1}(\mathbf{u}_{ref} + \mathbf{V}_{tot} \mathbf{q}_N; \boldsymbol{\mu}) \right\|_2^2.$$

This trajectory defines

$$\mathbf{q}_N^{lin}(t_k; \boldsymbol{\mu}) = \begin{bmatrix} \mathbf{q}^{lin}(t_k; \boldsymbol{\mu}) \\ \bar{\mathbf{q}}^{lin}(t_k; \boldsymbol{\mu}) \end{bmatrix}.$$

Case-2 oracle setting

The oracle high coefficients are prescribed as

$$\bar{\mathbf{q}}^{oracle}(t_k; \boldsymbol{\mu}) = \bar{\mathbf{q}}^{lin}(t_k; \boldsymbol{\mu}),$$

and only low coordinates are solved online. The trial state is

$$\begin{aligned} \mathbf{w}(\mathbf{q}, t_k; \boldsymbol{\mu}) &= \mathbf{u}_{ref} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \bar{\mathbf{q}}^{oracle}(t_k; \boldsymbol{\mu}), \\ \mathbf{q} &\in \mathbb{R}^n. \end{aligned}$$

The low-coordinate objective used in this report is the projected LSPG objective:

$$\begin{aligned} \mathbf{q}_{k+1}^{oracle,proj}(\boldsymbol{\mu}) &= \arg \min_{\mathbf{q} \in \mathbb{R}^n} \frac{1}{2} \left\| \mathbf{s}^{k+1}(\mathbf{q}; \boldsymbol{\mu}) \right\|_2^2, \\ \mathbf{s}^{k+1}(\mathbf{q}; \boldsymbol{\mu}) &= \left(\mathbf{J}^{k+1}(\mathbf{w}(\mathbf{q}, t_{k+1}; \boldsymbol{\mu}); \boldsymbol{\mu}) \mathbf{V}_{tot} \right)^T \\ &\quad \cdot \mathbf{r}^{k+1}(\mathbf{w}(\mathbf{q}, t_{k+1}; \boldsymbol{\mu}); \boldsymbol{\mu}). \end{aligned}$$

Oracle protocol. For the unperturbed oracle (0%), at each step $k+1$ we set

$$\mathbf{q}_{k+1}^{(0)} = \mathbf{q}_{low}^{lin}(t_{k+1}; \boldsymbol{\mu}), \quad \mathbf{w}_k = \mathbf{u}_{lin}(t_k; \boldsymbol{\mu}),$$

and solve the same projected reduced optimization. For Proposal 2 and Proposal 3, linear coefficients are taken in the same corrected basis used online. This defines the 0% oracle experiment as a direct reproduction test of the corresponding linear reduced trajectory.

4 Why high-mode errors contaminate low coordinates

Define

$$\mathbf{J}_L^{k+1} = \mathbf{J}^{k+1} \mathbf{V}, \quad \mathbf{J}_H^{k+1} = \mathbf{J}^{k+1} \bar{\mathbf{V}}.$$

For fixed high coordinates, the local low-coordinate objective is

$$\Phi_{k+1}(\mathbf{q}; \bar{\mathbf{q}}; \boldsymbol{\mu}) = \frac{1}{2} \left\| \mathbf{r}^{k+1}(\mathbf{u}_{ref} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \bar{\mathbf{q}}; \boldsymbol{\mu}) \right\|_{\mathbf{P}}^2,$$

with stationarity

$$\nabla_{\mathbf{q}} \Phi_{k+1} = (\mathbf{J}_L^{k+1})^T \mathbf{P} \mathbf{r}^{k+1} = \mathbf{0}.$$

If high coefficients are perturbed,

$$\bar{\mathbf{q}} \mapsto \bar{\mathbf{q}} + \delta \bar{\mathbf{q}},$$

then, after linearization,

$$\delta \mathbf{q} = - \left((\mathbf{J}_L^{k+1})^T \mathbf{P} \mathbf{J}_L^{k+1} \right)^{-1} \left((\mathbf{J}_L^{k+1})^T \mathbf{P} \mathbf{J}_H^{k+1} \right) \delta \bar{\mathbf{q}}.$$

Hence the transfer operator

$$\mathbf{T}^{k+1} = \left((\mathbf{J}_L^{k+1})^T \mathbf{P} \mathbf{J}_L^{k+1} \right)^{-1} \left((\mathbf{J}_L^{k+1})^T \mathbf{P} \mathbf{J}_H^{k+1} \right)$$

controls the amplification from high-mode errors to low-mode errors.

How to measure $\|\mathbf{T}\|$ and identify dangerous high-mode directions

The linearized relation

$$\delta \mathbf{q}^{k+1} \approx -\mathbf{T}^{k+1} \delta \bar{\mathbf{q}}^{k+1}$$

implies that the relevant gain is the induced operator norm

$$\|\mathbf{T}^{k+1}\|_2 = \sup_{\delta \bar{\mathbf{q}} \neq 0} \frac{\|\delta \mathbf{q}\|_2}{\|\delta \bar{\mathbf{q}}\|_2}.$$

Therefore, $\|\mathbf{T}^{k+1}\|_2$ is the worst-case local amplification from high-coordinate perturbations to low-coordinate perturbations.

In practice, we compute $\mathbf{T}^{k+1}$ at each time step from the linear reference trajectory:

$$\mathbf{T}^{k+1} \in \arg \min_{\mathbf{X} \in \mathbb{R}^{n \times \bar{n}}} \left\| \mathbf{J}_L^{k+1} \mathbf{X} - \mathbf{J}_H^{k+1} \right\|_{\mathbf{P}, F},$$

which is equivalent to

$$\left((\mathbf{J}_L^{k+1})^T \mathbf{P} \mathbf{J}_L^{k+1} \right) \mathbf{T}^{k+1} = (\mathbf{J}_L^{k+1})^T \mathbf{P} \mathbf{J}_H^{k+1},$$

with a small Tikhonov regularization in the $n \times n$ system when needed.

Now write the singular value decomposition

$$\mathbf{T}^{k+1} = \mathbf{U}^{k+1} \mathbf{\Sigma}^{k+1} (\mathbf{V}^{k+1})^T, \quad \mathbf{\Sigma}^{k+1} = \text{diag}(\sigma_1^{k+1}, \dots, \sigma_r^{k+1}),$$

where $r = \min(n, \bar{n})$ and

$$\sigma_1^{k+1} \geq \sigma_2^{k+1} \geq \dots \geq \sigma_r^{k+1} \geq 0.$$

Then:

- $\sigma_1^{k+1} = \|\mathbf{T}^{k+1}\|_2$: worst-case amplification factor.
- The right singular vector $\mathbf{v}_1^{k+1} \in \mathbb{R}^{\bar{n}}$ gives the high-mode perturbation direction that maximizes low-mode contamination.
- The corresponding low-mode response direction is $\mathbf{u}_1^{k+1} \in \mathbb{R}^n$.

For an actual perturbation $\delta \bar{\mathbf{q}}$, let

$$\delta \bar{\mathbf{q}} = \sum_{i=1}^r c_i \mathbf{v}_i, \quad c_i = (\mathbf{v}_i)^T \delta \bar{\mathbf{q}}.$$

Then

$$\delta \mathbf{q} \approx - \sum_{i=1}^r \sigma_i c_i \mathbf{u}_i,$$

and the squared low-error is decomposed as

$$\|\delta \mathbf{q}\|_2^2 \approx \sum_{i=1}^r \sigma_i^2 c_i^2.$$

This decomposition is the quantitative way to say “which high-mode directions are dangerous”: large σ_i with large projection c_i dominate contamination.

Hence, beyond reporting a single 1% perturbation test, the $\mathbf{T}$ -SVD analysis explains *why* contamination is mild or severe, and whether the observed error is aligned with intrinsically sensitive directions.

5 Three metric-based proposals

Proposal 1: weighted POD with physically consistent metric

Weighted POD is defined from

$$\mathbf{C} = \mathbf{S}^T \mathbf{W} \mathbf{S}.$$

The baseline choice is

$$\mathbf{W} = \mathbf{M}_{block} = \begin{bmatrix} \mathbf{M}_h & 0 \\ 0 & \mathbf{M}_h \end{bmatrix},$$

which corresponds to a discrete L^2 -type norm of the vector state.

To include LSPG sensitivity directly in the basis extraction, use

$$\mathbf{W}_{eff} = \varepsilon \mathbf{M}_{block} + \sum_s \omega_s \mathbf{J}_s^T \mathbf{P}_s \mathbf{J}_s, \quad \omega_s \geq 0, \quad \sum_s \omega_s = 1.$$

Proposal 2: modify only high modes

Keep $\mathbf{V}$ fixed and set

$$\tilde{\mathbf{V}} = \bar{\mathbf{V}} - \mathbf{V} \mathbf{K}, \quad \mathbf{K} = (\mathbf{H}_{LL} + \lambda \mathbf{I})^{-1} \mathbf{H}_{LH},$$

with

$$\mathbf{H}_{LL} = \mathbf{V}^T \mathbf{A} \mathbf{V}, \quad \mathbf{H}_{LH} = \mathbf{V}^T \mathbf{A} \bar{\mathbf{V}}, \quad \mathbf{A} = \sum_s \omega_s \mathbf{J}_s^T \mathbf{P}_s \mathbf{J}_s.$$

This enforces low-high orthogonality in metric $\mathbf{A}$ while preserving the low online subspace.

Proposal 3: modify only low modes

Keep $\bar{\mathbf{V}}$ fixed and set

$$\tilde{\mathbf{V}} = \mathbf{V} - \bar{\mathbf{V}} \mathbf{L}, \quad \mathbf{L} = (\mathbf{H}_{HH} + \lambda \mathbf{I})^{-1} \mathbf{H}_{HL},$$

with

$$\mathbf{H}_{HH} = \bar{\mathbf{V}}^T \mathbf{A} \bar{\mathbf{V}}, \quad \mathbf{H}_{HL} = \bar{\mathbf{V}}^T \mathbf{A} \mathbf{V}.$$

6 Numerical protocol at $\mu = (4.56, 0.019)$

Five basis families are compared:

- Euclidean POD basis,
- Proposal 1 with M_{block} ,
- Proposal 1 with W_{eff} ,
- Proposal 2 correction applied to Proposal 1 (W_{eff}),
- Proposal 3 correction applied to Proposal 1 (W_{eff}).

For each family, the following are evaluated:

1. Linear reduced reference trajectory.
2. Oracle test with 0% perturbation.
3. Oracle test with 1% perturbation of $\bar{\mathbf{q}}^{oracle}$.

For the Weff campaign reported here, Stage-1 and correction metrics are built with all available offline states:

$$\{(\mu_i, t_k)\}_{i=1, \dots, N_\mu, k=0, \dots, N_t},$$

without temporal subsampling and without random sample caps. The LSPG-averaged metric uses quadrature-consistent weights

$$\omega_{i,k} \propto w_i^\mu w_k^t, \quad w_i^\mu = \frac{1}{N_\mu},$$

with trapezoidal weights in time:

$$w_k^t = \begin{cases} \frac{\Delta t}{2}, & k = 0, \\ \Delta t, & 1 \leq k \leq N_t - 1, \\ \frac{\Delta t}{2}, & k = N_t. \end{cases}$$

After normalization,

$$\sum_{i=1}^{N_\mu} \sum_{k=0}^{N_t} \omega_{i,k} = 1.$$

The regularization term in W_{eff} is set by trace scaling:

$$\epsilon = \eta \frac{\text{tr}(C_J)}{\text{tr}(C_M)},$$

where $C_M = S^T M_{block} S$, $C_J = S^T \left(\sum_{i,k} \omega_{i,k} J_{i,k}^T P_{i,k} J_{i,k} \right) S$, and $\eta > 0$ is a dimensionless user parameter. This makes ϵM_{block} relative to the LSPG-sensitivity scale instead of using an absolute ϵ .

For cross-method comparison, n_{tot} is fixed to 151 modes in all branches. This isolates the effect of metric choice/correction from the effect of changing reduced dimension.

7 Results

Note on oracle protocol. All oracle tables below are obtained with the protocol defined in Section 2: the 0% oracle check uses the linear-reference previous state and step initialization for each basis family. Only the projected-residual oracle formulation is reported.

Definition of reported error columns

All trajectory errors are reported on the common time grid and in percent. Let

$$\begin{aligned} U^a &= [\mathbf{u}^a(t_0), \dots, \mathbf{u}^a(t_{N_t})], \\ U^{lin} &= [\mathbf{u}^{lin}(t_0), \dots, \mathbf{u}^{lin}(t_{N_t})], \\ U^{hdm} &= [\mathbf{u}^{hdm}(t_0), \dots, \mathbf{u}^{hdm}(t_{N_t})]. \end{aligned}$$

where a denotes the method/basis row of each table. The state errors used in the tables are

$$\text{rel. err. vs HDM}(\%) = 100 \frac{\|U^a - U^{hdm}\|_F}{\|U^{hdm}\|_F},$$

$$\text{rel. err. vs linear}(\%) = 100 \frac{\|U^a - U^{lin}\|_F}{\|U^{lin}\|_F}.$$

For the 1% oracle perturbation test, the same formulas are applied with perturbed trajectories $U^{a,\delta}$:

$$\text{rel. err. vs HDM (1\%)(\%)} = 100 \frac{\|U^{a,\delta} - U^{hdm}\|_F}{\|U^{hdm}\|_F},$$

$$\text{rel. err. vs linear (1\%)(\%)} = 100 \frac{\|U^{a,\delta} - U^{lin}\|_F}{\|U^{lin}\|_F}.$$

For reduced coordinates, with

$$Q_{low}^a = [\mathbf{q}_{low}^a(t_0), \dots, \mathbf{q}_{low}^a(t_{N_t})], \quad Q_{low}^{lin} = [\mathbf{q}_{low}^{lin}(t_0), \dots, \mathbf{q}_{low}^{lin}(t_{N_t})],$$

the low-coefficient errors are

$$\text{rel. low-coeff (0\%)(\%)} = 100 \frac{\|Q_{low}^a - Q_{low}^{lin}\|_F}{\|Q_{low}^{lin}\|_F},$$

$$\text{rel. low-coeff (1\%)(\%)} = 100 \frac{\|Q_{low}^{a,\delta} - Q_{low}^{lin}\|_F}{\|Q_{low}^{lin}\|_F}.$$

In Stage-1 diagnostics, the reported POD “energy lost” is

$$\text{energy lost} = 1 - \frac{\sum_{i=1}^{n_{keep}} \lambda_i}{\sum_{i=1}^{n_{avail}} \lambda_i},$$

where λ_i are eigenvalues of the corresponding correlation operator.

In the transfer-operator table, the drop column is

$$\text{drop in } \max_k \sigma_1(\mathbf{T}^k) \text{ vs Euclidean(\%)} = 100 \left(1 - \frac{\max_k \sigma_1(\mathbf{T}^k)^{(a)}}{\max_k \sigma_1(\mathbf{T}^k)^{(eucl)}} \right).$$

Stage-1 basis diagnostics

Basis	n_{keep}	n_{avail}	energy lost	metric orthogonality check
Euclidean	151	4509	9.97×10^{-5}	–
Proposal 1 (M_{block})	151	4509	9.97×10^{-5}	$\ V^T W V - I\ _F = 2.16 \times 10^{-14}$
Proposal 1 (W_{eff})	151	1508	1.17×10^{-4}	– (Gram check skipped in this run)

For Proposal 2 and Proposal 3 (built from Proposal 1 with W_{eff}):

$$\|H_{LH}\|_F : 2.34 \times 10^{-12} \rightarrow 2.34 \times 10^{-24},$$

$$\|H_{HL}\|_F : 2.34 \times 10^{-12} \rightarrow 2.34 \times 10^{-24}.$$

However, correction amplitudes are very small:

$$\|\mathbf{K}\|_F = 2.34 \times 10^{-12}, \quad \|\mathbf{L}\|_F = 2.34 \times 10^{-12}.$$

Linear reduced reference errors

Basis	rel. error vs HDM (%)	iterations	online time (s)
Euclidean	0.45	1503	1.06×10^3
Proposal 1 (M_{block})	0.45	1503	1.05×10^3
Proposal 1 (W_{eff})	0.46	1500	1.04×10^3
Proposal 2	0.46	1500	1.04×10^3
Proposal 3	0.46	1500	1.04×10^3

Oracle test, 0% perturbation

$$\bar{\mathbf{q}}^{oracle}(t; \boldsymbol{\mu}) = \bar{\mathbf{q}}^{lin}(t; \boldsymbol{\mu}).$$

Basis	rel. err. vs HDM (%)	rel. err. vs linear (%)	rel. low-coeff (%)
Euclidean	0.45	2.00×10^{-6}	4.71×10^{-6}
Proposal 1 (M_{block})	0.45	2.00×10^{-6}	4.71×10^{-6}
Proposal 1 (W_{eff})	0.46	1.77×10^{-6}	4.19×10^{-6}
Proposal 2	0.46	1.77×10^{-6}	4.19×10^{-6}
Proposal 3	0.46	1.77×10^{-6}	4.19×10^{-6}

Oracle perturbation tests (five-seed statistics at 1% and 2%)

To assess directional representativeness at fixed amplitude, we ran five independent random perturbation directions (seeds 11, 23, 42, 77, 101) for each basis family, at both perturbation levels 1% and 2%, for a total of 50 oracle-PG runs. Each perturbation is normalized to satisfy

$$\frac{\|\delta \bar{\mathbf{q}}\|_F}{\|\bar{\mathbf{q}}^{oracle}\|_F} \approx \alpha\%, \quad \alpha \in \{1, 2\}.$$

All entries are reported as mean $\pm$ standard deviation across the five seeds, in percent. In the table, labels are abbreviated as: E (Euclidean), $P1(M)$ (Proposal 1 with M_{block}), $P1(W)$ (Proposal 1 with W_{eff}), $P2$ (Proposal 2), $P3$ (Proposal 3).

Basis	perturbation level	err. vs linear (%)	err. vs HDM (%)	rel. low-coeff (%)
E	1%	$0.094 \pm 4.890\text{e-}4$	$0.459 \pm 1.120\text{e-}4$	$0.022 \pm 9.290\text{e-}3$
E	2%	$0.188 \pm 1.360\text{e-}3$	$0.487 \pm 3.940\text{e-}4$	0.061 ± 0.020
P1(M)	1%	$0.094 \pm 4.890\text{e-}4$	$0.459 \pm 1.120\text{e-}4$	$0.022 \pm 9.290\text{e-}3$
P1(M)	2%	$0.188 \pm 1.360\text{e-}3$	$0.487 \pm 3.940\text{e-}4$	0.061 ± 0.020
P1(W)	1%	$0.090 \pm 1.130\text{e-}4$	$0.468 \pm 1.160\text{e-}4$	$0.015 \pm 3.820\text{e-}3$
P1(W)	2%	$0.181 \pm 3.850\text{e-}4$	$0.493 \pm 2.160\text{e-}4$	$0.041 \pm 9.330\text{e-}3$
P2	1%	$0.090 \pm 1.130\text{e-}4$	$0.468 \pm 1.160\text{e-}4$	$0.015 \pm 3.820\text{e-}3$
P2	2%	$0.181 \pm 3.850\text{e-}4$	$0.493 \pm 2.160\text{e-}4$	$0.041 \pm 9.330\text{e-}3$
P3	1%	$0.090 \pm 1.130\text{e-}4$	$0.468 \pm 1.160\text{e-}4$	$0.015 \pm 3.820\text{e-}3$
P3	2%	$0.181 \pm 3.850\text{e-}4$	$0.493 \pm 2.160\text{e-}4$	$0.041 \pm 9.330\text{e-}3$

At 1%, the Weff family (Proposal 1 with W_{eff} , Proposal 2, Proposal 3) lowers the mean low-coordinate contamination from $2.18 \times 10^{-2}\%$ to $1.45 \times 10^{-2}\%$, i.e. 33.7% reduction relative to

Euclidean/ M_{block} . At 2%, the same reduction persists: $6.09 \times 10^{-2}\% \rightarrow 4.13 \times 10^{-2}\%$, i.e. 32.3% reduction. Across both families, doubling perturbation amplitude from 1% to 2% approximately doubles err. vs linear (factor ≈ 2.0), while rel. low-coeff grows superlinearly (factor ≈ 2.8).

Transfer-operator norm and SVD diagnostics

Using the linear reference trajectory, we evaluated $\mathbf{T}^k$ at each time step and extracted $\sigma_1(\mathbf{T}^k) = \|\mathbf{T}^k\|_2$. Here σ_1 denotes the largest *singular value* of $\mathbf{T}^k$, not an eigenvalue. This distinction is essential because $\mathbf{T}^k \in \mathbb{R}^{n \times \bar{n}}$ is rectangular. By definition of the induced Euclidean norm,

$$\|\mathbf{T}^k\|_2 = \sup_{\mathbf{x} \neq 0} \frac{\|\mathbf{T}^k \mathbf{x}\|_2}{\|\mathbf{x}\|_2} = \sigma_{\max}(\mathbf{T}^k) = \sigma_1(\mathbf{T}^k),$$

so σ_1 is the worst-case low-mode amplification factor for a unit perturbation in high-coordinate space.

The corresponding right singular vector $\mathbf{v}_1^k \in \mathbb{R}^{\bar{n}}$ identifies the high-mode direction that maximizes contamination at step k , and the left singular vector $\mathbf{u}_1^k \in \mathbb{R}^n$ gives the associated low-mode response direction.

Basis	$\max_k \sigma_1(\mathbf{T}^k)$	mean $\sigma_1(\mathbf{T}^k)$	p95 $\sigma_1(\mathbf{T}^k)$	drop in $\max_k \sigma_1$ vs Euclidean (%)
Euclidean	3.47×10^{-2}	2.84×10^{-2}	3.38×10^{-2}	0.00
Proposal 1 (M_{block})	3.47×10^{-2}	2.84×10^{-2}	3.38×10^{-2}	0.00
Proposal 1 (W_{eff})	2.06×10^{-2}	1.46×10^{-2}	1.97×10^{-2}	40.80
Proposal 2	2.06×10^{-2}	1.46×10^{-2}	1.97×10^{-2}	40.80
Proposal 3	2.06×10^{-2}	1.46×10^{-2}	1.97×10^{-2}	40.80

Hence, in this run, the measurable reduction in the low/high contamination operator is produced by Proposal 1 (W_{eff}) relative to Euclidean/ M_{block} , while Proposal 2 and Proposal 3 do not add further reduction beyond that Weff baseline. An auxiliary fit diagnostic,

$$\frac{\|\mathbf{J}_L \mathbf{T} - \mathbf{J}_H\|_F}{\|\mathbf{J}_H\|_F},$$

is close to 1 for all branches in this case, meaning $\mathbf{J}_H$ is largely transverse to $\text{Range}(\mathbf{J}_L)$; therefore $\mathbf{T}$ captures the projected contamination channel, not a full reconstruction of high-block Jacobian directions.

Dominant right-singular directions at σ_1 -peak. The leading entries of the right singular vector $\mathbf{v}_1^k$ were extracted at the time step where $\sigma_1(\mathbf{T}^k)$ is maximal for each branch. Indices below are reported with the implementation indexing convention (0-based). For clarity, if an entry is reported as $j : a$, it means

$$(\mathbf{v}_1^k)_j = a,$$

where j refers to the j -th coordinate in the secondary-coefficient space $\bar{\mathbf{q}} \in \mathbb{R}^{\bar{n}}$. Hence the same entry corresponds to full reduced index $n + j$ in

$$\mathbf{q}_N = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}.$$

Equivalently, in state space, this component acts through column j of $\bar{\mathbf{V}}$, i.e. through the mode $\bar{\mathbf{V}}_{(:,j)}$.

Branch	$\arg \max_k \sigma_1(\mathbf{T}^k)$	t_k	dominant entries of $\mathbf{v}_1^k$ (index:value)
Euclidean / $P1(M)$	$k = 40$	2.0	6 : $-2.201\text{e-}1$, 5 : $2.192\text{e-}1$, 7 : $2.010\text{e-}1$, 4 : $-1.796\text{e-}1$
Weff / $P2$ / $P3$	$k = 42$	2.1	5 : $2.539\text{e-}1$, 12 : $-2.427\text{e-}1$, 4 : $-2.383\text{e-}1$, 6 : $-2.355\text{e-}1$

This confirms that the dominant contamination directions differ between the $E/P1(M)$ branches and the Weff family.

8 Interpretation of the observed behavior

(i) Euclidean and M_{block} branches

At this mesh and test setting, both branches are numerically almost identical: same retained dimension, same energy loss, same linear error, and near-identical oracle behavior. This is consistent with the fact that the discrete mass weighting is close to a scaled Euclidean weighting on a structured grid.

(ii) Effect of W_{eff}

The enriched metric modifies the snapshot correlation operator and changes the reduced basis geometry. This is visible in

- slightly different linear reference error,
- lower $\|\mathbf{T}\|_2$ levels than Euclidean/ M_{block} in this campaign.

(iii) Why Proposal 2 and Proposal 3 overlap here

In this run, computed correction operators are extremely small: $\|K\|_F \sim 10^{-12}$, $\|L\|_F \sim 10^{-12}$. Consequently, Proposal 2 and Proposal 3 remain almost indistinguishable from the uncorrected Weff basis, hence nearly identical numerical outputs.

(iv) Unperturbed oracle reproduction in all branches

Under the oracle protocol above, the 0% mismatch is near machine precision for all five basis families:

$$\text{rel. err. vs linear} \in [1.77 \times 10^{-6}, 2.00 \times 10^{-6}],$$

for the projected-residual solve.

The Weff-family cross products remain non-negligible in Euclidean coordinates,

$$\|V^T \bar{V}\|_F \approx \begin{cases} 2.24 \times 10^{-15} & \text{Euclidean,} \\ 3.01 \times 10^{-14} & \text{Proposal 1 } (M_{block}), \\ 2.81 \times 10^{-2} & \text{Weff / Proposal 2 / Proposal 3.} \end{cases}$$

and this does not prevent near-exact reproduction in the unperturbed oracle test.

9 Oracle protocol and $\mathbf{T}$ diagnostics

The same oracle protocol is applied in every branch:

1. the oracle high coefficients are taken from the corresponding linear reduced trajectory in the same basis;
2. at each time step, the previous state and reduced initialization are taken from that linear trajectory;
3. the same projected-residual reduced objective is minimized in all branches.

For the perturbation experiment, only the oracle high block is modified so that

$$\frac{\|\delta \bar{\mathbf{q}}\|_F}{\|\bar{\mathbf{q}}^{oracle}\|_F} \approx \alpha\%,$$

$\alpha \in \{1, 2\}$ in this report, while keeping the reduced basis and the online reduced formulation unchanged.

Transfer-operator diagnostics are then evaluated along the same linear reference trajectory, reporting $\max_t \sigma_1(\mathbf{T}^t)$, mean and p95 values, and the fit quality of $\mathbf{J}_H \approx \mathbf{J}_L \mathbf{T}$. These diagnostics connect the measured contamination in oracle perturbation tests with the predicted linearized sensitivity directions.

10 Conclusions and immediate next experiments

1. The M_{block} -weighted branch reproduces the Euclidean baseline almost exactly in this setting.
2. The W_{eff} branch changes basis geometry and modifies both linear and oracle responses.
3. At 0%, all branches reproduce the corresponding linear reference up to $\mathcal{O}(10^{-6})\%$ in state and reduced coordinates.
4. Proposal 1 (W_{eff}) reduces $\max_k \|\mathbf{T}^k\|_2$ from 3.47×10^{-2} to 2.06×10^{-2} , i.e. about 40.80% reduction versus Euclidean/ M_{block} .
5. In five-seed perturbation tests, the Weff family lowers low-coordinate contamination relative to Euclidean/ M_{block} at both amplitudes: 33.7% reduction at 1% and 32.3% at 2%.
6. In this dataset, Proposal 2 and Proposal 3 do not reduce $\|\mathbf{T}\|_2$ further beyond the Weff baseline, consistent with $\|K\|_F, \|L\|_F \sim 10^{-12}$.